The two types of spores are microspores and megaspores. **Microspores** grow into a male gametophyte, and **megaspores** grow into a female gametophyte (see Fig. 23.18).

Check Your Progress 23.3

1. Name two features of lycophytes that increase their ability to survive on land.
2. Explain how xylem contributes to an upright body plant.
3. Define the terms *homosporous* and *heterosporous*.

23.4 Evolution of Pteridophytes: Megaphylls

Learning Outcomes

Upon completion of this section, you should be able to

1. Identify three types of pteridophytes.
2. Compare and contrast microphylls and megaphylls.
3. Identify the components of the ferns' life cycle.

Pteridophytes is a broad term used to describe a group of seedless vascular plants that includes ferns and their allies, the horsetails and whisk ferns. Both pteridophytes and the seed plants have megaphylls. **Megaphylls** are broad leaves with several strands of vascular tissue. Figure 23.12*a* shows the difference between microphylls and megaphylls, and Figure 23.12*b* shows how megaphylls could have evolved.

Megaphylls, which evolved independently from the evolution of microphylls, allow plants to efficiently collect solar energy, leading to the production of more food and the possibility of producing more offspring than plants without megaphylls. Therefore, the evolution of megaphylls made plants more fit. Recall that fitness, in an evolutionary sense, is judged by the number of living offspring an organism produces relative to others of its own kind.

The pteridophytes, like the lycophytes, were dominant from the late Devonian period through the Carboniferous period. Today, the lycophytes are quite small, but some of the extinct relatives of today's club mosses were 35 m tall and dominated the Carboniferous swamps. The horsetails (around 18 m tall) and ancient tree ferns (8 m or more in height) also contributed significantly to the great swamp forests of the time (see the following Evolution feature, "Carboniferous Forests").

Horsetails

Today, **horsetails** consist of one genus, *Equisetum,* and approximately 15 species of distinct seedless vascular plants. Most horsetails inhabit wet, marshy environments around the globe. About 299 MYA, horsetails were dominant plants and grew as large as

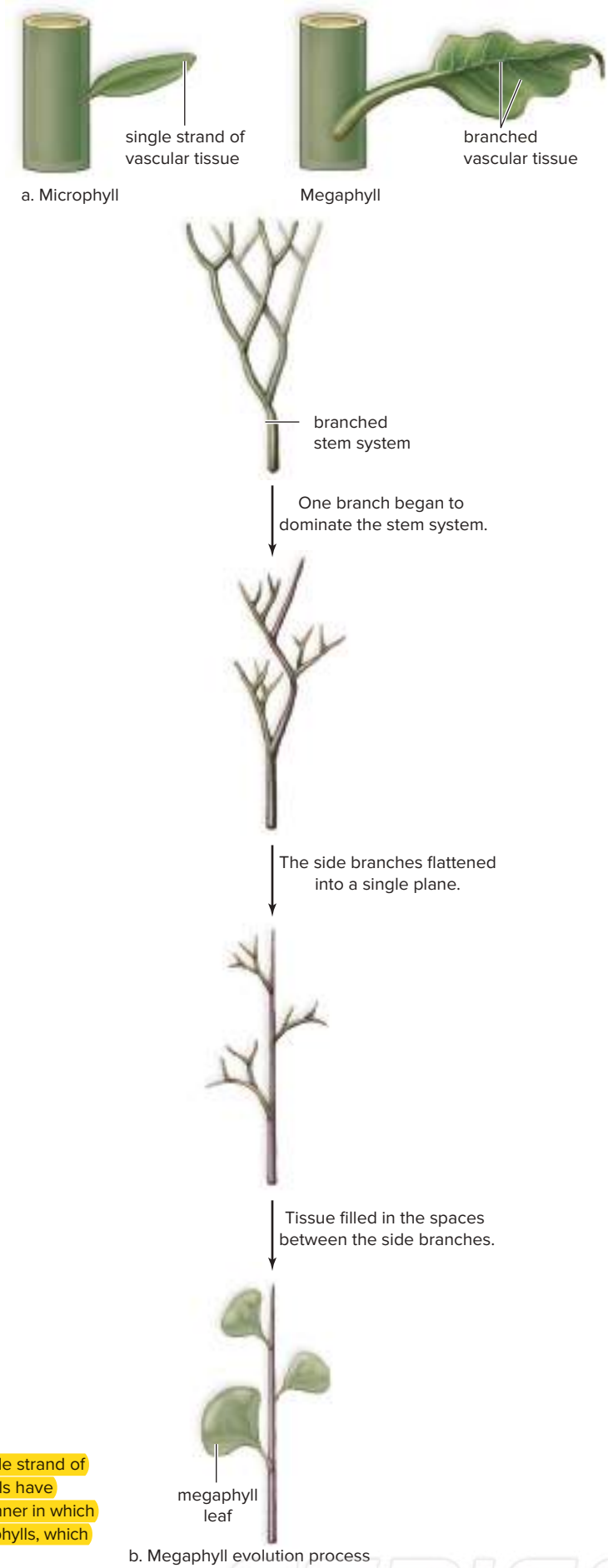


Figure 23.12 Microphylls and megaphylls. **a.** Microphylls have a single strand of vascular tissue, which explains why they are quite narrow. In contrast, megaphylls have several branches of vascular tissue and are broader. **b.** These steps show the manner in which megaphylls may have evolved. All vascular plants, except lycophytes, bear megaphylls, which can gather more sunlight and produce more organic food than microphylls.